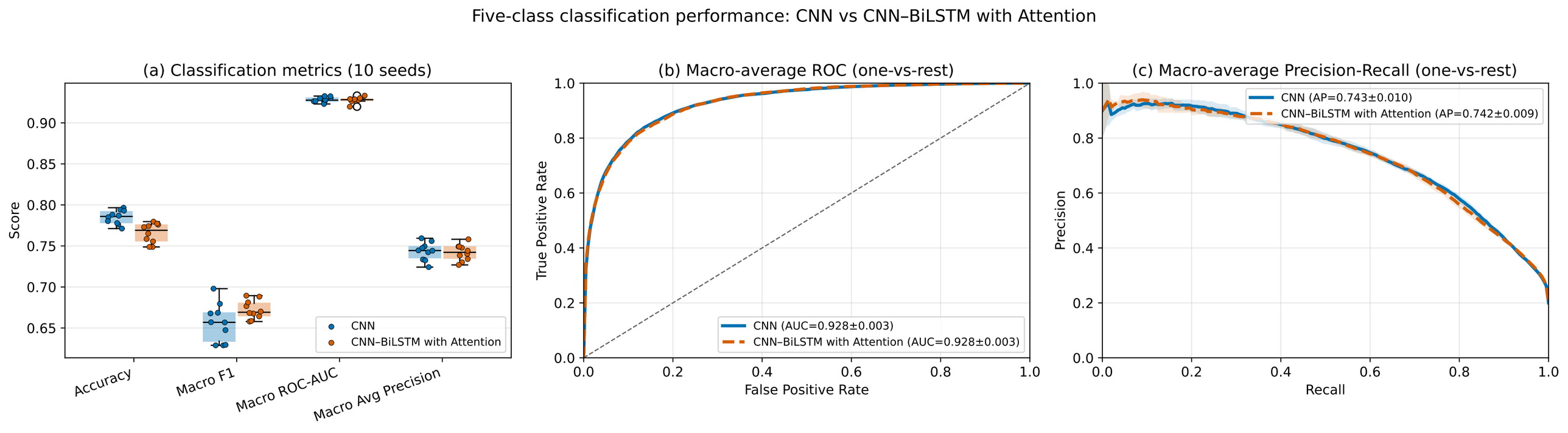
Five-class classification performance: CNN vs CNN–BiLSTM with Attention
(a) Classification metrics (10 seeds)
Score
0.90
0.85
0.80
0.75
0.70
0.65
CNN
CNN–BiLSTM with Attention
Accuracy
Macro F1
Macro ROC-AUC
Macro Avg Precision
(b) Macro-average ROC (one-vs-rest)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
CNN (AUC=0.928±0.003)
CNN–BiLSTM with Attention (AUC=0.928±0.003)
(c) Macro-average Precision-Recall (one-vs-rest)
Precision
Recall
CNN (AP=0.743±0.010)
CNN–BiLSTM with Attention (AP=0.742±0.009)